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Zn-doped aerogel for Ni^{2+} adsorption (Zn-A-Ni) and reuse of Zn-A-Ni to create Zn, Ni-co-doped carbon aerogel for applications in adsorption and energy storage

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A B S T R A C T

In order to solve the problem of heavy metal pollution, in this study, Zn-doped aerogel (Zn-A) was used as a material to remove Ni^{2+} from wastewater. Zn-A was synthesized from sodium alginate and nipa palm shell-derived cellulose via the sol–gel method combined with freeze-drying. Zn-A has the ability to adsorb Ni^{2+} up to 194.2 mg/g (Zn-A-Ni). After adsorbing Ni^{2+}, Zn-A-Ni was pyrolyzed to form Zn, Ni-co-doped carbon aerogel (Zn-CA-Ni), which has high potential for manufacturing electrodes in supercapacitors (specific capacitance reaches 124.0F/g) and crystal violet treatment (adsorption capacity reaches 38.7 mg/g). Furthermore, Zn-A and Zn-CA-Ni were characterized through modern methods: Scanning electron microscope, energy dispersive X-ray, Fourier-transform infrared spectroscopy, X-ray diffraction analysis, and Nitrogen adsorption–desorption isotherm. The ability to adsorb Ni^{2+} of Zn-A and adsorb crystal violet of Zn-CA-Ni was determined through ultraviolet–visible spectroscopy measurement. In addition, the electrochemical properties of Zn-CA-Ni were also analyzed through cyclic voltammetry, galvanostatic charge–discharge, and electrochemical impedance spectroscopy.

Introduction

Water resources always play an important role in the lives of humans in particular and microorganisms on Earth in general. However, the amount of clean water is decreasing day by day. According to World Vision statistics, the problem of chronic water shortages is threatening nearly 700 million people around the world and this number could increase to 1.8 billion by 2025 (according to The United Nations Water) [1]. Depending on the depth compared to the ground, water is divided into groundwater and surface water, with many different causes of pollution. In particular, the surface layer is easiest to observe when many places become dead rivers, the water becomes black, has an odor or floating garbage, etc. The main cause of this phenomenon is domestic wastewater and especially water sources from manufacturing plants, leading to surface water containing large amounts of heavy metals, colorants, antibiotics, garbage, etc. In this study, the subject mentioned is heavy metals and colorant pollution in wastewater. In recent years, heavy metal pollution has significantly increased due to global population growth and industrial advancements. The harmful effects of these metals on human health have been extensively substantiated, highlighting their significant and far-reaching implications. These toxic substances can be found in a range of items, including electronic devices, machinery, and everyday commodities [2,3]. Furthermore, industrial wastewater, urban sewage, and mining waste, particularly within the

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electroplating and metal fabrication industries, are major contributors to the release of heavy metals into the environment. Consequently, these pollutants directly infiltrate the human body and animal organisms via a variety of pathways, posing immediate, and chronic poisoning risks, damaging effects on vital organs such as the liver, kidneys, and intestines, and the possibility of the development of conditions such as anemia, cancer, even mortality, etc. [4,5].

Noteworthy heavy metals commonly observed in significant concentrations within aquatic ecosystems include arsenic (As), cadmium (Cd), copper (Cu), mercury (Hg), manganese (Mn), lead (Pb), nickel (Ni), etc. The content of heavy metals in wastewater is mentioned in Table 1 [6]. Depending on the environment and geographical location, the amount of metal in water changes, but the negative impact it has on human health and vitality is huge.

To address the stringent environmental and health standards, numerous methods have been proposed for the treatment of water contaminated with heavy metals. With previous researches, several treatment techniques were investigated such as membrane filtration, microbial bioremediation, advanced oxidation process, carbon nanotechnology, electrochemical methods, chemical precipitation, solvent evaporation, ion exchange, adsorption, etc. Among them, chemical precipitation, membrane filtration techniques, and adsorbents are commonly used methods [7]. While chemical precipitation and membrane filtration techniques have been employed, adsorption methods are currently acknowledged as highly efficient and economically viable approaches, facilitating enhanced removal of heavy metals. With the method of using adsorbents, two operating mechanisms are mentioned: Physical and chemical adsorption. For physical adsorption, metals are adsorbed onto the material based on molecular size and Van der Waals forces [7]. Meanwhile, chemical adsorption occurs when the adsorbent undergoes a chemical reaction (ion exchange, precipitation, etc. (Fig. 1) [8].

Nowadays, adsorbents are becoming more and more diverse, such as capillary materials, nanomaterials, carbon materials, etc. Most of these materials can adsorb metal ions thanks to their porous structure, high specific surface area, increased ability to contact adsorbed ions, etc. Accordingly, airgel is also a porous material and has the potential to adsorb heavy metal ions.

Aerogel is a porous material with a three-dimensional (3D) structure, high porosity (90 %), lightweight, and low density (0.0011–0.5 g/cm^3). Among the various adsorbents, aerogel exhibits great promise as a material characterized by exceptional properties [9–11]. Several studies have shown the adsorption capacity of aerogel for dyes, metal ions, etc., as shown in Table 2.

In that way, the aerogel is doped with metals or modified to increase porosity and surface area, increasing adsorption efficiency. Therefore, a Zn-doped aerogel was synthesized in this research because Zn is a transition metal with high ionic mobility and chemical compatibility. To synthesize aerogel, gelation, hydrogel, and freeze-drying methods have been applied. During the gel formation process, the cross-linking agent sodium alginate (SA) is used in combination with group II or III cations in an ion exchange reaction with Na+, creating a fixed gel system. In the process of creating aerogel, Zn^{2+} not only acts as a doping agent but also as a network connector, creating a stable gel system to consolidate the 3D structure of the aerogel. Additionally, Zn-doped aerogel acts as a precursor for the synthesis of ZnO-doped carbon aerogel. When aerogel incorporates metals, it transforms into metals-doped carbon aerogel, which finds application in energy storage. To adapt to the requirements of contemporary and future energy storage systems and devices, supercapacitors emerge as the optimal solution to address energy storage challenges [16,17].

Table 1
Contents of heavy metals in wastewater.

No.	Heavy metals	Contents (mg/kg)	WHO allowed permissible limits (mg/Kg)
1	Cu	28–1150	0.05
2	Cr	10–136	0.05
3	Ni	9–262	0.07
4	Pb	0–79	0.05
5	As	12.1–41.6	0.05
6	Hg	0.67–19.50	0.1
7	Cd	0.21–2.77	0.003

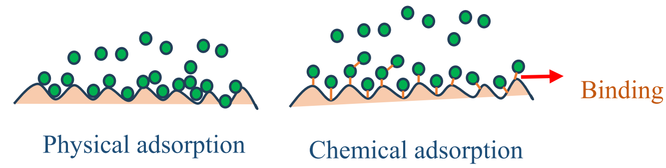


Fig. 1. Physical and chemical adsorption.

Table 2
Some studies on the adsorption capacity of aerogel.

No.	Materials	Applications	Ref.
1	Tannin-reduced graphene aerogel (TRGA)	The maximum Pb(II) adsorption capacity of TRGA reached 803.84 mg/g and Cd(II) is 395.80 mg/g at pH 4.	[12]
2	Elastic polyethyleneimine-modified carboxymethyl chitosan aerogel	Cu (II) adsorption capacity reached 175.56 mg/g.	[13]
3	Composite aerogels from graphene oxide and chitosan	The maximum adsorption amounts of MO and MB were calculated to be 543.4 and 110.9 mg/g at 25 °C	[14]
4	Polypyrrole (PPy)-modified natural eggplant (PPy@eggplant) aerogel	Cr^{6+} adsorption capacity of 27.83 mg/g.	[15]

In this study, Ni was selected as a typical heavy metal to investigate heavy metal adsorption. Ni^{2+} is a substance located in the Earth's crust, an ingredient in stainless steel with pure Ni being a shiny, silvery-white metal. Therefore, Ni is very often used for surface plating to create an external decorative layer. However, with too much Ni, human health will be adversely affected, such as: Increasing the likelihood of developing lung cancer, laryngeal cancer, nose cancer, heart disorders, etc. Therefore, the treatment of Ni in the aquatic environment is very important. Since then, aerogel has been investigated as a potential Ni^{2+} adsorbent [18–20]. Besides, Zn-doped aerogel absorbed Ni^{2+} can form Zn, Ni-co-doped carbon aerogel through pyrolysis.

The ZnO-doped aerogel had the ability to adsorb Ni and became a precursor for carbon aerogel (CA) synthesis to expand its application to the electrochemical field and dye adsorption. However, typical aerogels have usually been synthesized from chemical precursors such as resin, formaldehyde resin, etc. This affects the environment and is difficult to handle after use. Since then, research on replacing chemical precursors with cellulose from biomass. The biomass used in this study was nipa palm shell. Nipa palm is a fruit tree distributed in southern Vietnam, often growing along rivers. When harvested, only the flesh in the middle of the fruit is used for drinking water or canned food, and the outer shell is considered biomass waste. Nipa palm shell (NS) is known as a biomass by-product with large production in Vietnam, rich in cellulose (over 50 %), from which cellulose is easily separated from the shell and synthesized aerogel materials [21,22]. During hydrogel formation, Zn^{2+} solution was used, this increased the reducing agent, generating ZnO with particles [23,24]. The hydrogel was made from cellulose nipa palm shells, sodium alginate as the cross-linking agent, Zn^{2+} as the network coupling agent, and as a doping agent, the hydrogel was then sublimed to produce CA [25].

Carbon aerogel is known as an advanced material with special characteristics: High porosity, large specific surface area, and diverse 3D porous structure, and thanks to the carbonization process, it forms CA with more than 90 % carbon [26,27]. Therefore, CA becomes a potential material for making electrodes in supercapacitors. Supercapacitors have the ability to store energy with a specific capacitance about 3–5 times higher than conventional batteries and capacitors. The structure of the supercapacitor is described in Fig. 2: Two electrodes from the material, electrolyte solution [28]. Thanks to the compatibility between charges, a double electrical layer is formed, thereby increasing the ability to store energy. The energy storage ability of CA is determined through the electrochemical properties of the three-electrode system, including the reference electrode (calomel), reference electrode (Pt), and working electrode (CA/glassy carbon electrode (CA/GCE)).

In addition, CA also has the ability to adsorb organic dyes such as activated carbon. Therefore, the ability of CA to adsorb organic colorants was also investigated in this study with crystal violet (CVt) chosen as a typical colorant to investigate [29,30]. CVt is a purple compound that is widely used in textile dyeing, medicine, etc. It has a ring structure, so it is very durable in the natural environment. From there, CVt causes great harm to the environment and humans.

When the Zn-doped aerogel (Zn-A) adsorbed Ni (Zn-A-Ni), Ni is considered to be the doping agent in the pyrolysis of CA into Zn, Ni-co-doped CA (Zn-CA-Ni). Thus, in this study, aerogel was synthesized from cellulose nipa palm shell, zinc acetate salt, and sodium alginate through cross-linking combined freeze-drying methods. Then, the aerogel was used as the Ni^{2+} adsorbent. Utilizing Ni^{2+}-adsorbed aerogel to synthesize CA by pyrolysis. Aerogel and CA were characterized through material analysis methods: Scanning electron microscope (SEM), energy dispersive X-ray (EDX), Fourier-transform infrared spectroscopy (FTIR), X-ray diffraction analysis (XRD), and Nitrogen adsorption–desorption isotherm by Brunauer-Emmett-Teller (BET). Then Zn-CA-Ni was applied in energy storage and CVt adsorption. The Ni^{2+} adsorption capacity of the Zn-A and the ability to adsorb CVt of Zn-CA-Ni were determined through the ultraviolet–visible spectroscopy method (UV–Vis). The energy storage capacity of Zn-CA-Ni was demonstrated by its electrochemical properties with cyclic voltammetry (CV), galvanostatic charge–discharge (GCD), and electrochemical impedance spectroscopy (EIS) results.

Materials and method

Materials

Nipa palm shells were collected from street vendors (District 10, Ho Chi Minh City) and synthesized in our previous research [22]. The chemicals used included: Sodium hydroxide (NaOH, solid), acetic acid (CH_3COOH, liquid 98 %), formic acid (HCOOH, liquid 98 %), sodium alginate (SA) ($NaC_6H_7O_6$, solid), zinc chloride ($ZnCl_2$, solid), crystal violet ($C_{25}N_3H_{30}Cl$, solid), and hydrogen peroxide (H_2O_2, liquid 98 %) were provided by Xilong, China. Sodium sulfate (Na_2SO_4, solid), Nafion®, and ethanol (C_2H_5OH, 99 %), were supplied by Sigma-Aldrich, Germany. De-ionized water (DI water) was used throughout all experiments.

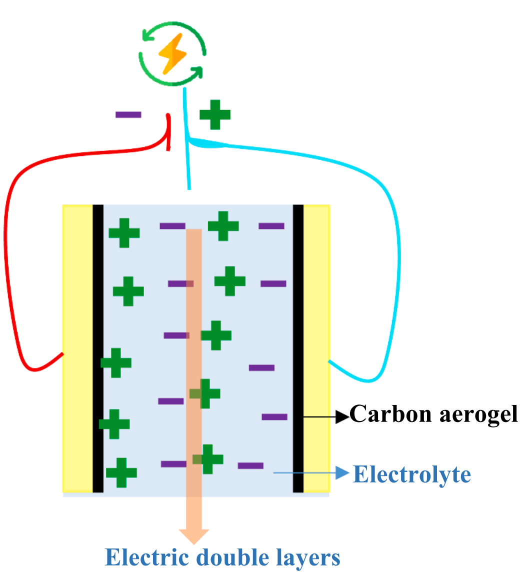


Fig. 2. Mechanism of supercapacitor.

Methods

Preparation of Zn, Ni-co-doped carbon aerogel

Zn-doped aerogel and Zn-CA-Ni were synthesized through a simple process as shown in Scheme 1.

Synthesis of the ZnO-doped aerogel (Zn-A)

Sodium alginate was weighed for 1 g and then put in 50 mL of DI water, then stirred at 60 °C for 30 min to form a transparent gel [23]. Besides, nipa palm-derived cellulose was synthesized in previous research [22]. Following that, cellulose was weighed and put into the SA solution with the masses noted in Table 3. The gel system was stirred for 15 min and at 60 °C to ensure the SA formed even crosslinkages with the cellulose. After which, 50 mL of the Zn^{2+} 10 % solution was put into the gel system, stirred, and let to settle the gel system in 10 min to stabilize. Then, it was sonicated, incubated, and then frozen to form hydrogel (Zn-H). Zn-H was freeze-dried to form Zn-A

Synthesis of Zn, Ni-co-doped carbon aerogel

Aerogel was investigated as Ni^{2+} adsorbent. After adsorbing Ni^{2+}, Zn-A-Ni underwent pyrolysis to form Zn-CA-Ni in the conditions listed in Table 4.

From there, the suitable temperature to synthesize Zn-CA-Ni for energy storage applications was given. The CA at the same pyrolysis condition without Ni was examined for comparison.

Characterization

SEM images and EDX results were collected using JMS-IT 200, Jeol, Japan with an acceleration voltage of 10 kV, magnification × 10000, resolution 512383, and dwell time of 0.20 ms. Raman spectra were recorded with labram 300, jobin yvon instrument with the measuring range of 0–4500 cm^{-1}, measuring medium in a mixture of he and ne inert gases. FTIR spectra were determined via the KBr pellets method at a resolution of 0.2 cm^{-1}, the accuracy of the spectral range 0.1 % T, and the surveyed wavelength range 400–4000 cm^{-1}. Samples were measured using an XRD D8 advance of Bruker, germany. Cu–Kα radiation source had the wavelength $\lambda = 1.5406$ nm; scanning angle $2\theta = 5$-$80°$. The specific surface area was determined via the N_2 adsorption–desorption curve at a temperature of 77.35 K and a pressure of 756 mmHg.

Moreover, the density and porosity of aerogel was determined by Equations (1) and (2).

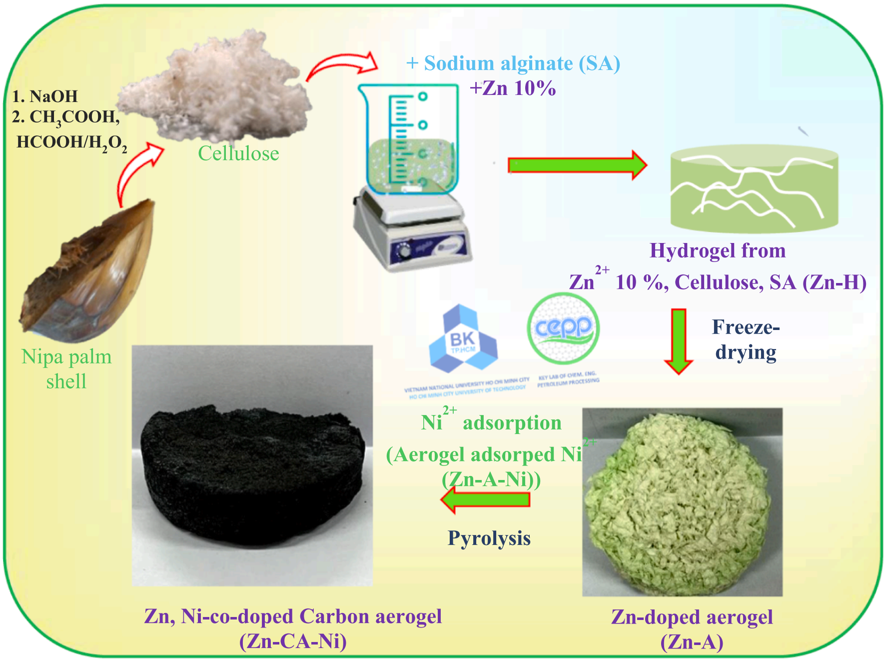
$$d = \frac{m}{V}(g/cm^3) \tag{1}$$

$$\rho = \left(1 - \frac{d_o}{d}\right) \times 100\% \tag{2}$$

where d is the density of aerogel (g/cm^3) and is calculated by mass (m, g) and volume (V, cm^3) of this aerogel. From that, porosity (ρ, %) was determined through d, d_o, with d_o being the density of cellulose [31].

Nickel (II) adsorption capacity of the Zn-A

The synthesized Zn-A has the ability to adsorb Ni^{2+} heavy metal ions and the adsorption capacity was determined through UV–Vis spectroscopy by a machine with the maximum absorption wavelength of Ni^{2+} at 395 nm. The mass of Zn-A was fixed at 15 mg in 20 mL of Ni^{2+} solution,



Scheme 1. Schematic diagram of synthesis Zn-doped aerogel and Zn, Ni-co-doped-carbon aerogel.

Table 3
Mass of cellulose in Zn-A.

Number	Samples	Mass of cellulose	Volume of Zn^{2+} 10 %	Content of SA
1	Zn-A1	3		
2	Zn-A2	5		
3	Zn-A3	7	50 mL	1 %
4	Zn-A4	10		
5	Zn-A5	15		

Table 4
Pyrolysis temperature of aerogel to synthesize Zn-CA-Ni.

Number	Samples	Temperature (°C)
1	Zn-CA1-Ni	600
2	Zn-CA2-Ni	700
3	Zn-CA3-Ni	800

affecting the factors of time on the adsorption capacity of the investigated materials. Ni^{2+} adsorption capacity was calculated according to the Equation (3).

$$q_t = \frac{(C_o - C_t) \times V}{m} (\mathrm{mg/g}) \tag{3}$$

where q_t is the adsorption capacity of Ni^{2+} at time t (s); C_o and C_t (ppm) are concentrations of Ni^{2+} solution before and after adsorption, respectively. V is the volume of Ni^{2+} solution fixed at 20 mL, and m is the mass of material added (fixed at 10 mg of material/20 mL of Ni^{2+} solution).

Besides that, the Ni^{2+} adsorption process was investigated through pseudo-first and second-order kinetic models with Equations (4) and (5), respectively.

$$\ln(q_e - q_t) = \ln(q_e) - k_1 t \tag{4}$$

$$\frac{t}{q_t} = \frac{1}{k_2 {q_e}^2} + \frac{1}{q_e} t \tag{5}$$

where q_e is the adsorption capacities of the material at equilibrium (g/g), respectively. k_1 and k_2 are the kinetic constants of pseudo-first and second-order models, respectively.

Electrochemical measurement of Zn-CA-Ni

The electrochemical properties were determined by the CHI660C electrochemical meter through CV, GCD, and EIS methods. The Zn-CA and Zn-CA that have undergone Ni^{2+} adsorption (Zn-CA-Ni) were finely ground, mixed with carbon black, and dispersed in ethanol by ultrasonic waves to form a suspension A. Mixture A was coated on the glassy carbon surface with a volume of 10 µL, using 0.5 % Nafion® to increase adhesion, and fix the electrode surface. This glassy carbon electrode serves as the working electrode in a three-electrode system [32]. From CV results, EIS calculates specific capacitance and conductivity through Equations (7) and (8).

$$C_s = \frac{\int_{V_1}^{V_2} I(V) dV}{m v \Delta V} \tag{7}$$

$$\sigma = \frac{L}{RA} \tag{8}$$

where C_s (F/g) is the specific capacitance, I/m (A/g) is the applied current density, Δt (s) is the discharge time(s), ΔV(V) is the maximum potential window for cell discharge, and m is the mass of electrodes present on the electrode surface (g). σ (Ohm^{-1}) is the electrical conductivity of the material, L (cm) is the electrode thickness, R (Ohm) is the resistance, and A (cm^2) is the working area of the electrode.

Crystal violet adsorption capacity of Zn-CA-Ni

The CVt adsorption capacity of Zn-CA-Ni was investigated similarly to the Ni adsorption capacity of Zn-A using the UV–Vis method as presented in section 2.2.3.

Results and discussion

Characterization

The morphological characteristics of the materials (Zn-A2, Zn-CA1, Zn-A1-Ni, and Zn-CA1-Ni) were determined through the utilization of

SEM imagery, as exemplified in Figs. 3 and 4. In the context of Zn-A2, it becomes evident that a membranous stratum enveloping cellulose fibers is discernible. This phenomenon arises from the gelation induced by SA after cellulose immersion, thereby facilitating the intercalation of cellulose fibers and SA, leading to their alternation in a juxtaposed manner (Fig. 3a, b). The cellulose fibers exhibit non-uniform dimensions. It can be observed that, through the intermittent interweaving of cellulose fibers, porous voids are formed to retain these fibers. Furthermore, upon increasing the magnification (Fig. 3c), the presence of a coating layer of Zn and SA onto the cellulose fibers becomes discernible, exhibiting distinct regions. Following the absorption of Ni (Fig. 3d, f), it is evident that the aerogel maintains its porous structure, with the distinct presence of cellulose fibers (Fig. 3d). In Fig. 3e, f, a large area surrounding the cellulose fibers, originating from SA, is apparent. However, this layer is continuous, rather than discrete patches as depicted in Fig. 3c. This phenomenon can be explained by the adsorption of Ni, where Ni^{2+} adheres to the material's surface, uniformly filling the porous voids, consequently connecting the regions and reducing the discreteness.

Following thermal decomposition, significant differences in the structure of Zn-CA and Zn-CA-Ni became apparent in contrast to the aerogel. As depicted in Fig. 4a, b, there is clear evidence of substantial contraction in the dimensions of the cellulose fibers. These fibers display varying lengths and diameters, measuring less than 5 µm. However, in Fig. 4a, there is an evident fracture among the cellulose fibers, indicative of the partial influence of thermal decomposition on the material's structure. Upon the impregnation of Ni into the aerogel, Ni also becomes affixed to the material as an impurity phase through the method of immersion impregnation, thereby giving rise to the composite material with heterogeneous phases, namely Zn-CA-Ni. Normally, ZnO particles adopt a spherical configuration, and in Fig. 4c, the presence of spherical particles and rod-like structures akin to fibers in Zn-CA-Ni showcases the distinctive morphological presence of Ni. This occurrence serves to enhance crystal lattice defects and augments potential applications for the material.

Besides, the crystal plane of Zn-CA-Ni is revealed through the XRD pattern in Fig. 5. Around 2 Theta gets 23°, a broad and diffuse peak is evident, which is a distinctive signal of graphite C. Additionally, there is a sharp peak at 76° characteristic of the (220) crystal plane of NiO. Given the relatively low Ni content, few distinctive crystal planes of NiO are present. ZnO, with a considerable amount of impurity phases and introduced during the hydrogel formation stage, displays multiple distinctive peaks at 31, 32, 47, 56, 62, 67, 68, 69, and 73° corresponding to the (110), (002), (101), (110), (103), (200), (112), (201), and (004) crystal planes, respectively [10,33]. This confirms the certain incorporation of ZnO impurity particles into the CA structure with diverse crystalline networks.

The elemental composition of Zn-A2, Zn-A2-Ni, Zn-CA, and Zn-CA-Ni is presented in Fig. 6 and Table 5. The primary elemental constituents of Zn-A2 include C, O, and Zn with respective concentrations of 35.75; 39.16; and 9.99 %. In the case of aerogel, the carbon content, originating from carbohydrates, is relatively low, reaching only 35 %. Following the Ni adsorption, Zn-A2-Ni displays an additional presence of Ni, contributing to the elemental composition with a content of 2.69 %. This outcome underscores the potential for impurity incorporation through the adsorption process.

Following the thermal decomposition process, Zn-CA exhibits a high carbon content of 74.42 % and a reduced oxygen content (from 39.16 to 7.28 %), demonstrating the complete occurrence of thermal decomposition. After adsorption and thermal treatment, the nickel content within Zn-A2-Ni reaches 1.57 %. After adsorption and pyrolysis, the C and Ni contents decreased. This can be explained through the synthesis of Zn-CA-Ni, this material has two co-doped metals. However, Zn is introduced during the cross-linking stage, so it is quite durable. After pyrolysis, there is a slight decrease in Zn content from 19.72 to 18.27 %. For Ni mixed through impregnation, the Ni content is quite low and the link between Ni and the carbon base is unstable and easily lost during the pyrolysis process, so the amount of Ni decreases sharply. In addition, because the structure of Zn-A when investigating Ni adsorption changed, leading to the loss of C from the original aerogel after pyrolysis, the C content also decreased [33,34].

The distribution of elements on the surface of the material is depicted through the mapping image in Fig. 6. This reveals an uneven distribution of elements across the surface of the material.

The characteristic functional groups of Zn-A2, Zn-CA1, and Zn-CA1-Ni are elucidated through FTIR results (Fig. 7). For NS-cell, peaks at 3300, 2900, 1760, 1635, 1373, and 1060 cm^{-1} correspond to the functional groups – OH, –CH–, –C – O, C – C, C – H, and C – O, respectively, which are indicative of cellulose [18]. Upon the creation of Zn-A2 aerogel, cellulose-specific functional groups are present, along with an additional peak at 700 cm^{-1} characteristics of – Zn. In the case of Zn-A1-Ni, a peak at 609 cm^{-1} indicative of Ni is observed. However, after thermal decomposition, Zn-CA1 and Zn-CA1-Ni exhibit only the C

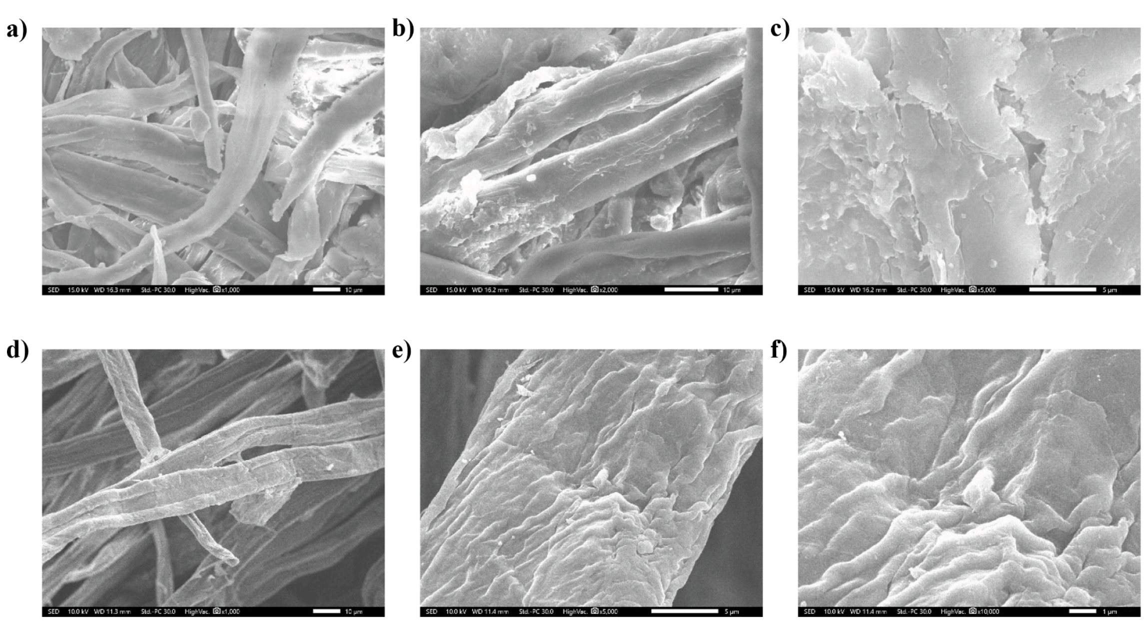


Fig. 3. SEM images of Zn-A2 (a-c) and Zn-A2-Ni (d-f).

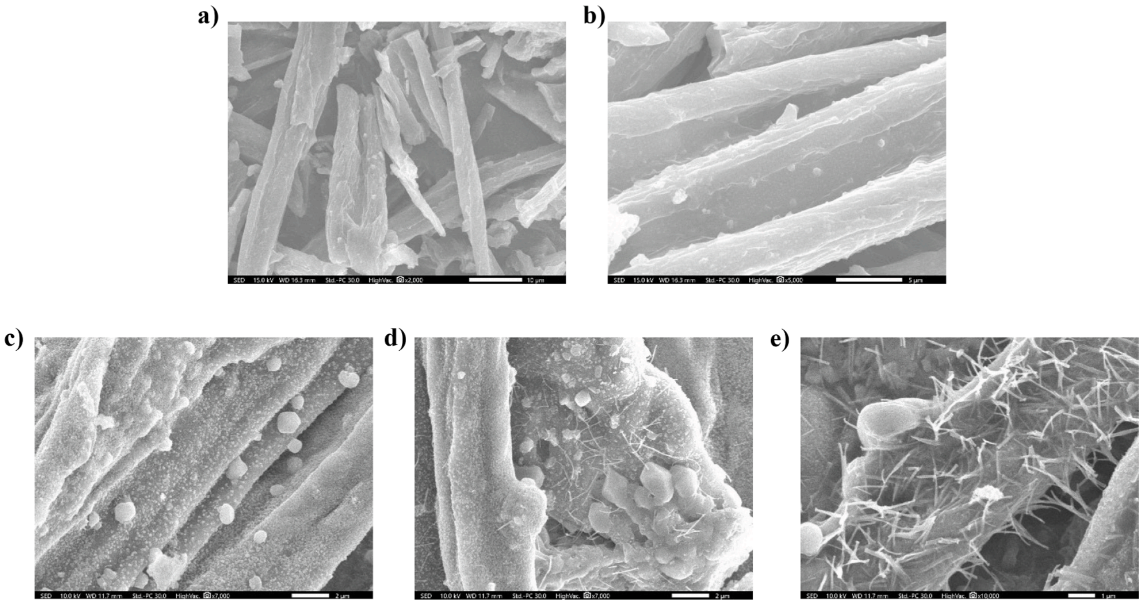


Fig. 4. SEM images of Zn-CA1 (a-b) and Zn-CA1-Ni (c-e).

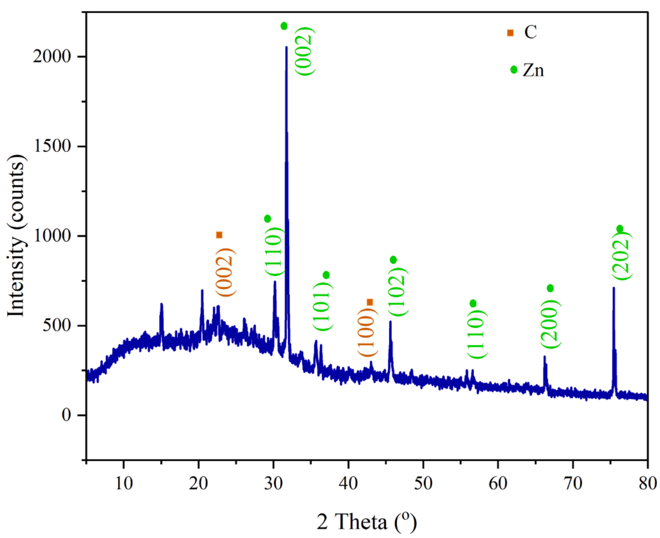


Fig. 5. XRD pattern of Zn-CA-2.

– C peak at 1600 cm^{-1}, and the baseline is nearly linear, lacking distinct signals of functional groups found in cellulose and aerogel [18,35].

To show the pore size of Zn-CA-Ni, Zn-CA1-Ni was chosen as a representative sample to investigate nitrogen adsorption–desorption. The pore size of Zn-CA1-Ni is illustrated in Fig. 8. The average pore size of the material measures 3.5 nm (Fig. 8b). In Fig. 8a, the N_2 adsorption–desorption curve of Zn-CA1-Ni is shown as type IV, with a clear phase lag, and the pore size is about micro-meso, with meso being the main size. The specific surface area of the material reaches only 29.33 m^2/g, which may be influenced by the Ni^{2+} adsorption process, partially occluding the pores and accordingly leading to a reduction in the specific surface area. Ni is doped through the impregnation process, so it partly covers the pores, thereby reducing the specific surface area. Although the specific surface area is low, materials with the presence of Zn and Ni (transition metals) have high electron mobility, thereby expanding their applications in adsorption and electrochemistry.

Nickel (II) adsoption capacity of Zn-A

The process of Ni^{2+} adsorption is compared among Zn-A with different contents of cellulose in aerogel, as shown in Fig. 9a. This analysis demonstrates the material's capability to adsorb Ni^{2+}, achieving levels as high as 194.2 mg/g, with Zn-A2 exhibiting the maximum nickel adsorption capacity. Meanwhile, Zn-A1, Zn-A3, Zn-A4 and Zn-A5 achieved Ni^{2+} adsorption capacities of 19.55, 68.95, 37.39, and 21.76 mg/g, respectively. The difference in Ni^{2+} adsorption capacity of Zn-A is explained based on the different cellulose content in the sample, which leads to different densities and porosities (Table S1 in supporting information file (SI file)). Accordingly, Zn-A2 has the lowest density (0.062 g/cm^3) and highest porosity (95.85 %). When the density is low and the porosity is high, it shows that the internal structure of Zn-A2 is porous, Ni^{2+} ions easily diffuse and interact from the surface to the inside of the material. Increasing the interaction area of Ni will achieve a higher adsorption capacity [30,36].

In addition, when the cellulose content is low (3 %) corresponding to Zn-A1, the porous structure of the aerogel is not durable enough and easily collapses. When the cellulose content is high, it is easy to cause overload and form a gel system instead of creating cross-links and pores. Consequently, Zn-A2 is selected for investigating the influence of time on the adsorption capacity. As depicted in Fig. 9b, the equilibrium state of the nickel adsorption process is reached at 120 min, enabling the determination of the kinetics through two models illustrated in Fig. 9c, d.

The linear lines corresponding to the pseudo-first-order (Fig. 9a) and second-order (Fig. 9d) kinetic adsorption models are shown. Accordingly, the correlation constant R^2 of the pseudo-second-order kinetic model is higher (0.859), thereby showing that the Ni adsorption process of Zn-A is more suitable for the pseudo-second-order model.

Thus, Zn-A2 was a suitable sample to adsorb Ni^{2+}, thereby pyrolyzing to make Zn-CA-Ni. With the presence of two metals (Zn and Ni) and the porous structure of CA, Zn-CA-Ni is a potential material for electrochemical applications.

Electrochemical properties of Zn-CA-Ni

The electrochemical characteristics of Zn-CA-Ni are elucidated in the findings presented in Fig. 10 and Table 6. As evident from Fig. 10a, the cyclic voltammetry (CV) curves exhibit a quasi-rectangular shape with

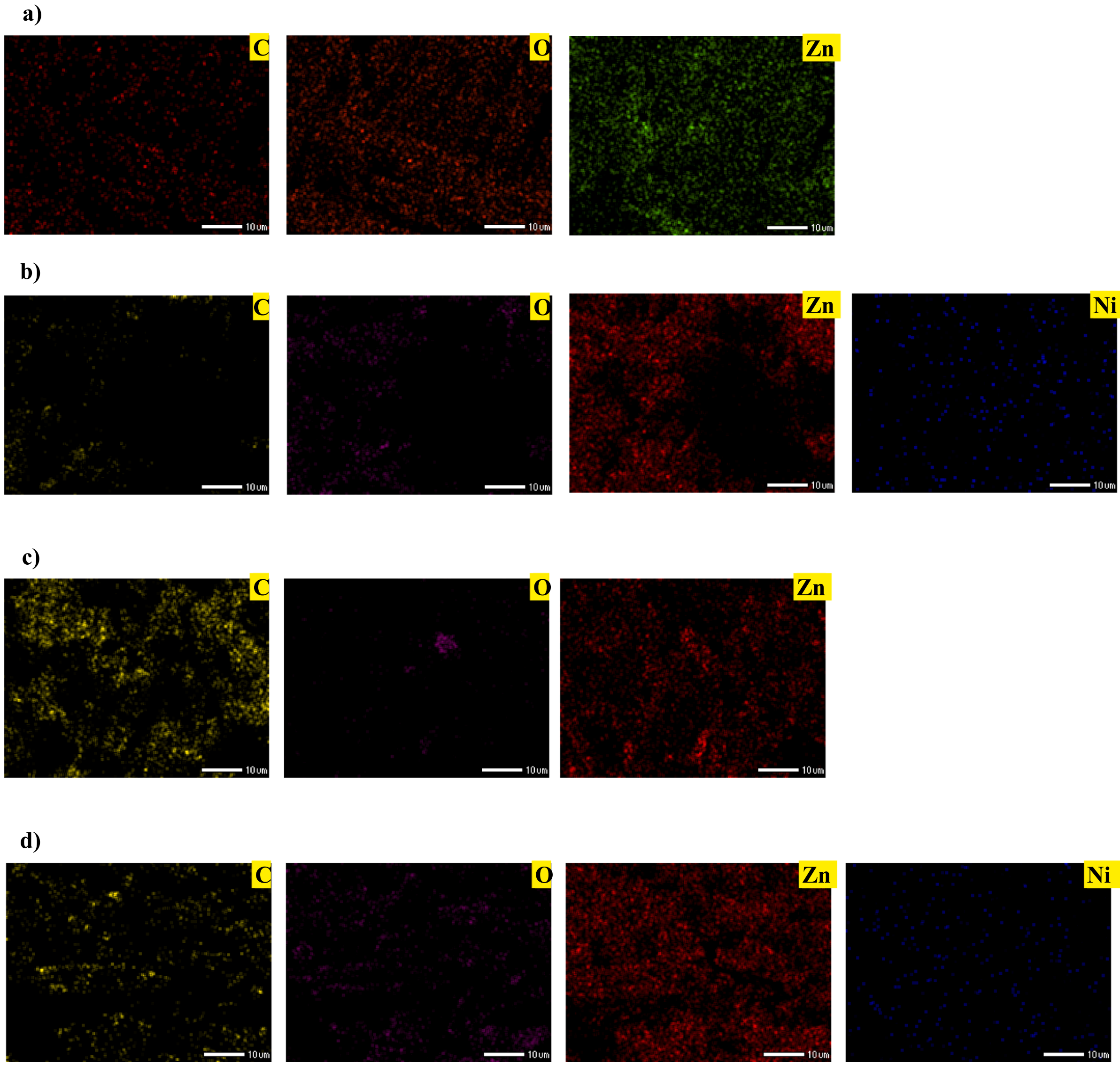


Fig. 6. EDX mapping images of (a) Zn-A2, (b) Zn-A2-Ni, (c) Zn-CA1, and (d) Zn-CA1-Ni.

Table 5
Percent of mass in Zn-A2 and Zn-CA2 before and after Ni^{2+} adsorption.

Number	Samples	Mass %			
		C	O	Zn	Ni
1	Zn-A2	35.75 ± 0.12	39.16 ± 0.18	9.99 ± 0.18	
2	Zn-A2-Ni	35.10 ± 0.16	15.24 ± 0.13	19.72 ± 0.24	2.69 ± 0.76
3	Zn-CA1	74.42 ± 0.18	7.28 ± 0.11	6.52 ± 0.14	
4	Zn-CA1-Ni	37.53 ± 0.15	16.75 ± 0.13	18.27 ± 0.21	1.57 ± 0.63

an absence of oxidation–reduction signals. The specific capacitance of Zn-CA1-Ni is observed to be higher (124.0F/g), as indicated in the statistical analysis in Table 6. Correspondingly, Zn-CA1-Ni got low resistances of 399.4 Ohm and high conductivity (0.042 Ohm^{-1}). This phenomenon can be readily explained by the presence of Ni as a transitional metal, which enhances electron mobility, thereby elevating the conductivity and energy storage capacity of the material.

Moreover, Zn-CA-Ni is utilized to investigate the durability of a

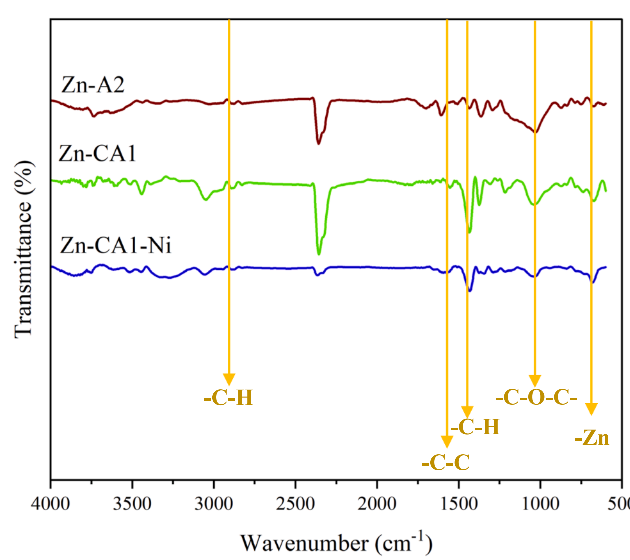


Fig. 7. FTIR spectra of Zn-A2, Zn-CA1, and Zn-CA1-Ni.

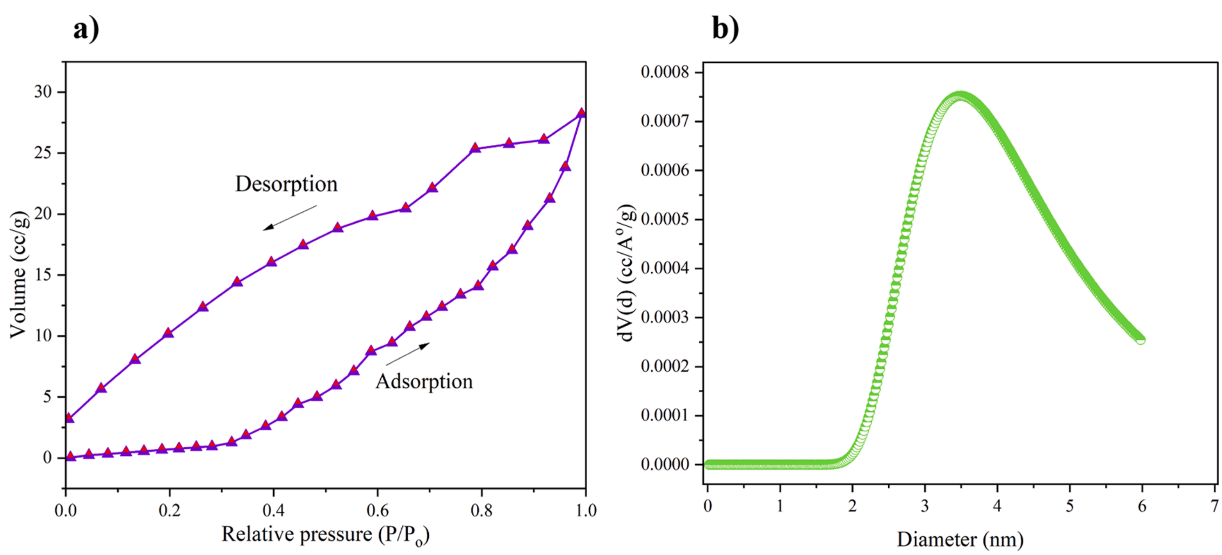


Fig. 8. (a) Nitrogen isotherm adsorption–desorption curve and (b) pore size distribution in Zn-CA1-Ni.

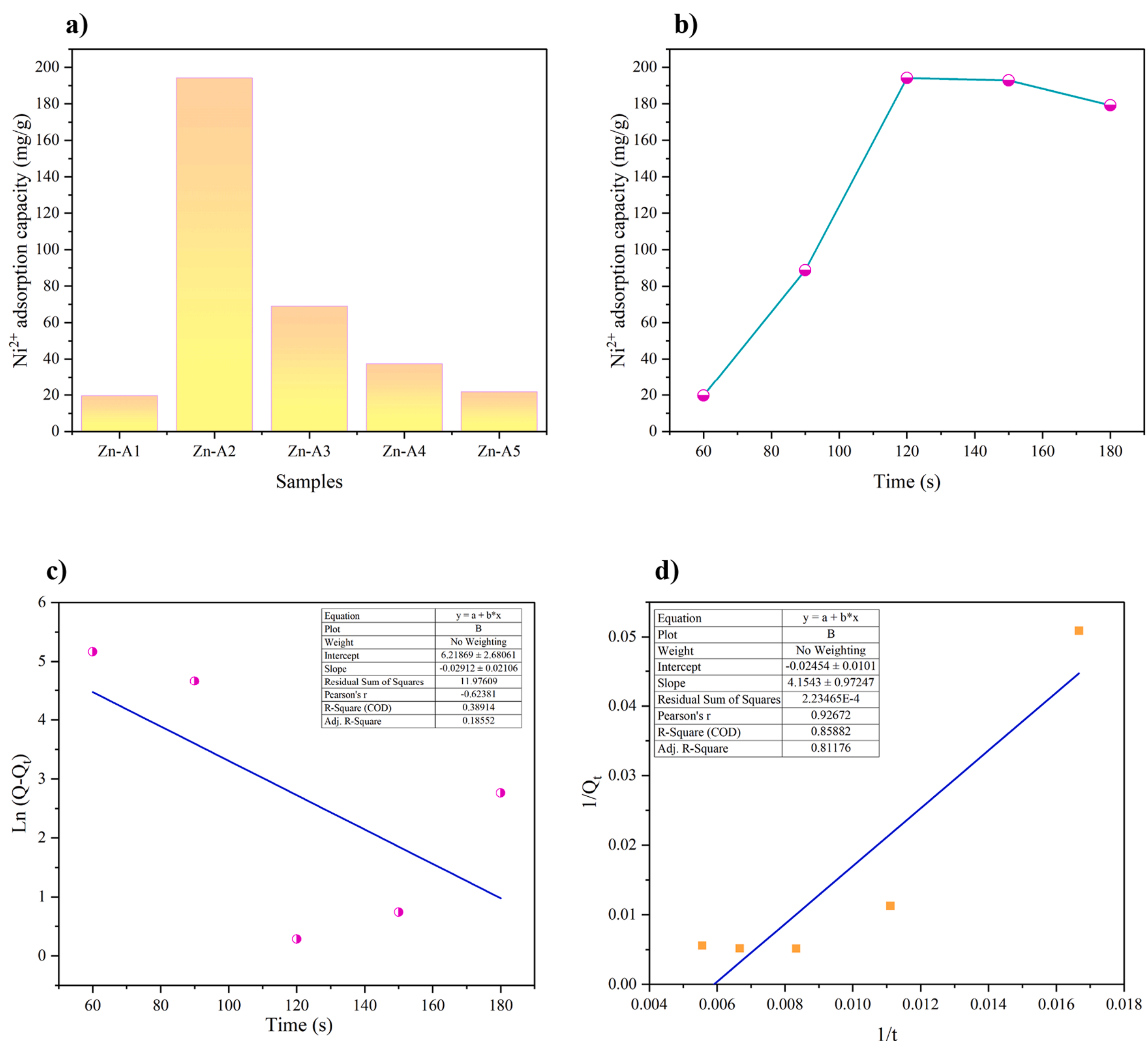


Fig. 9. (a) Ni^{2+} adsorption capacity of Zn-A with differences in mass of cellulose*, (b) Effect of time to Ni^{2+} adsorption capacity of Zn-A2**, (c) pseudo-first, and (d) second-order kinetic models Ni^{2+} adsorption of Zn-A2. (Experimental conditions. (*) 15 mg Zn-A, 20 mL Ni^{2+} 200 ppm, 120 min (**) 15 mg Zn-A-2, 20 mL Ni^{2+} 200 ppm).

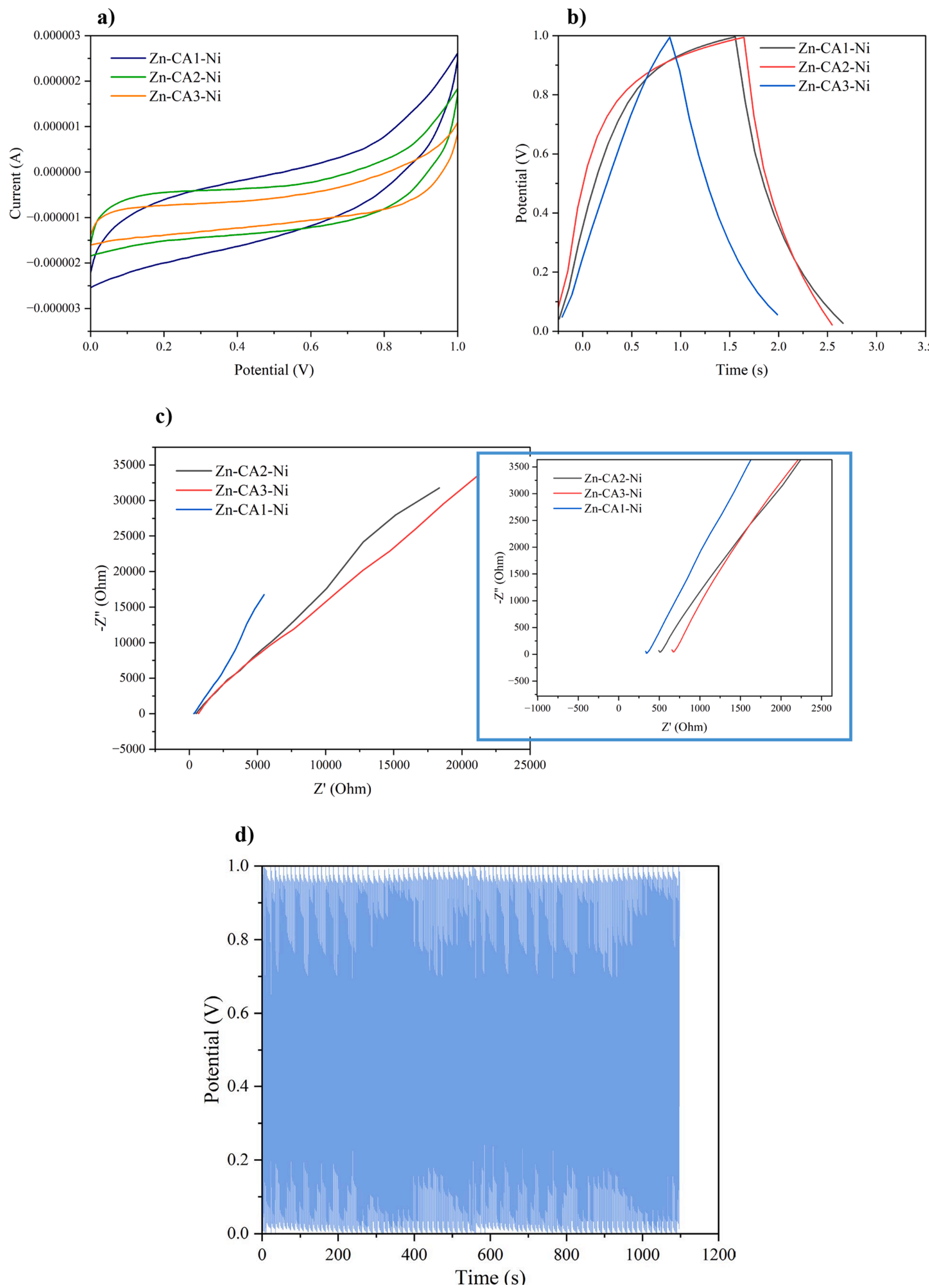


Fig. 10. (a) CV curves at 20 mV/s, (b) GCD curves at 2 A/g, (c) EIS results of Zn-CA-Ni, and (d) GCD curves of Zn-CA1-Ni after 1000 cycles.

supercapacitor system through the evaluation of GCD outcomes. The graph depicted in Fig. 10b, d, illustrates a recurring charging and discharging cycle of the supercapacitor system composed of Zn-CA-Ni in a triangular configuration, requiring only 2 s for completion. Upon analyzing 1000 cycles, the system's stability is observed to remain intact.

In this study, Zn-A showed the ability to adsorb Ni in water while helping to synthesize metal-doped CA. Therefore, to clearly show the

Table 6
Electrochemical parameters of Zn-CA-Ni.

Number	Samples	Specific capacity (F/g)	R (Ohm)	Conductivity (Ohm^{-1})
1	Zn-CA1-Ni	124.0	339.4	0.042
2	Zn-CA2-Ni	98.9	526.1	0.027
3	Zn-CA3-Ni	64.5	700.7	0.020

effect of Ni, the Zn-CA sample was determined electrochemically, as shown in Figure S1 and Table S2. Accordingly, Zn-CA also has a stable specific capacitance (102F/g) but is still lower than Zn-CA-Ni under the same synthesis conditions.

Like activated carbon, Zn-CA and Zn-CA-Ni also have the ability to adsorb organic dyes.

Crystal violet adsorption capacity of Zn-CA-Ni

The CVt adsorption capacity of Zn-CA and Zn-CA1-Ni was investigated for 150 min, as shown in Fig. 11a. Thereby, the time for the adsorption process to reach saturation is about 120 min with the adsorption capacity reaching 38.0 mg/g (Zn-CA) and 38.7 mg/g (Zn-CA1-Ni). So, Zn-CA1-Ni still shows wider applicability when Ni is present on a carbon base. From there, the adsorption kinetics of Zn-CA1-Ni was investigated, as shown in Fig. 11b, c. Accordingly, the pseudo-second-order adsorption kinetic model is more suitable for the CVt adsorption process for Zn-CA1-Ni (R^2 reaches 0.89).

The above findings indicate that aerogel and carbon aerogel mixed with metals have high application potential. Besides, utilizing aerogel as a heavy metal adsorbent and reusing aerogel after adsorption to synthesize CA shows high efficiency in adsorption and energy storage. The applicability of Zn-CA1-Ni is compared with some previous studies in Table 7.

Through Table 7, it can be seen that Zn-A is a potential material for removing heavy metals from waste-water and Zn-CA1-Ni is considered capable of adsorbing CVt and fabricating electrodes in supercapacitors, which shows that Zn-CA1-Ni has potentially in many fields. Although its application performance is not highly appreciated, Zn-CA1-Ni is considered a potential multi-application material, in addition to adsorbing Ni^{2+}, with an aerogel porous structure that can adsorb heavy

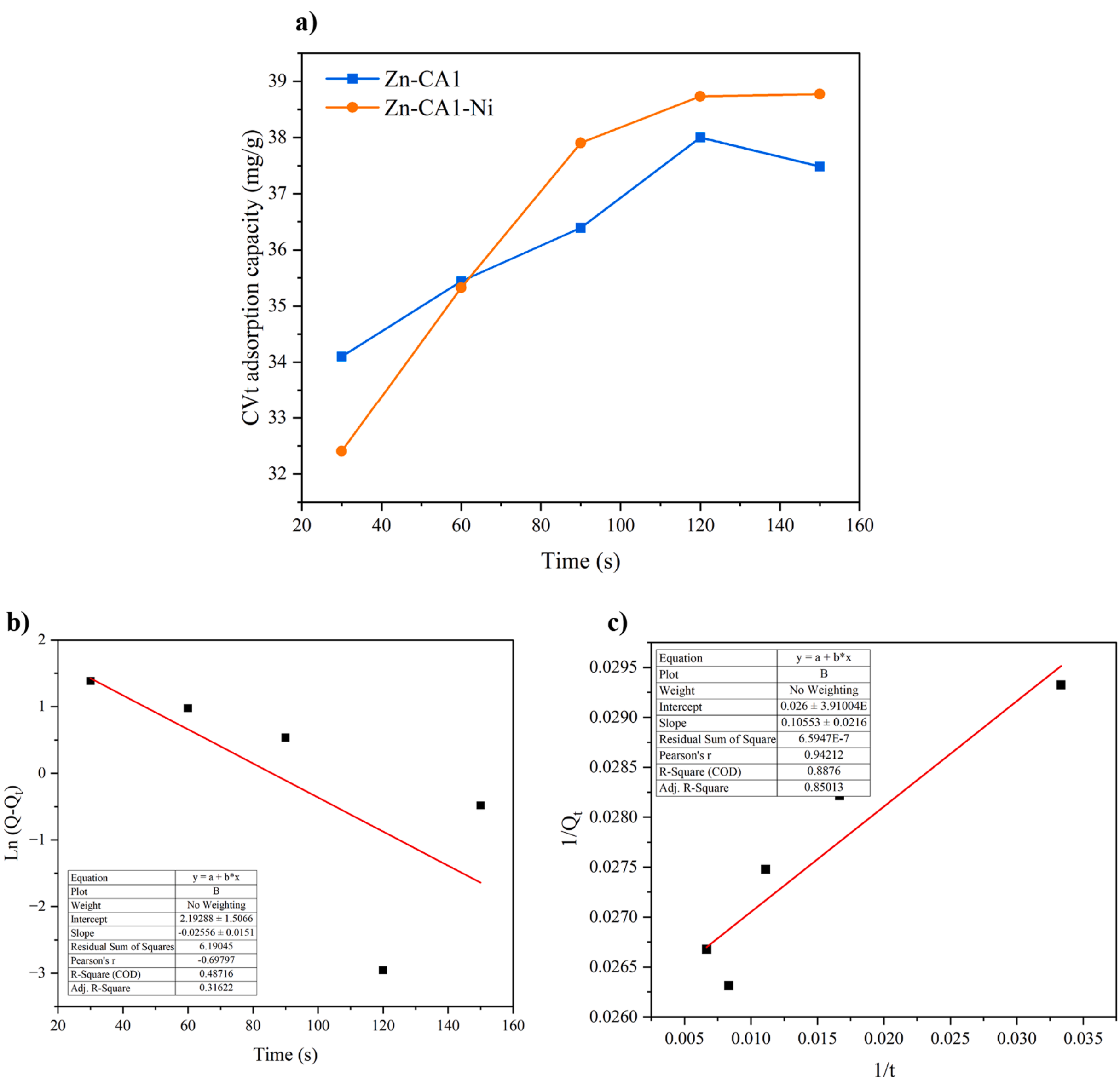


Fig. 11. (a) CVt adsorption capacity of Zn-CA and Zn-CA1-Ni, (b) pseudo-first, and (c) second-order kinetic models CVt^{+} adsorption of Zn-CA1-Ni.

Table 7
Some previous studies about carbon aerogels in adsorption and storage energy.

No.	Materials	Adsorption capacity	Application in energy storage	Ref.
1	Hierarchical porous carbon aerogel from liquefied wood		Specific capacitance of 154.15Fg^{-1} at 0.5Ag^{-1}	[37]
2	Magnetic ferric alginate carbon aerogel (FeACA)	The adsorption capacities of FeACA for methylene blue, Congo red, and oil red O were 123.7, 141.1, and 65.7 mg g^{-1}, respectively		[38]
3	Biochar-loaded carbon aerogels,	Adsorption capacity of 205.07 mg.g^{-1} at 25 °C for Pb(II), 105.56 mg.g^{-1} for Cd(II), and 137.89 mg.g^{-1} for Zn(II)		[39]
4	Zn-CA1-Ni	Zn-A has the ability to adsorb Ni up to 194.2 mg.g^{-1}; Zn-CA-Ni also adsorbs crystal violet pigment with an adsorption capacity of 38.7 mg.g^{-1}.	Specific capacitance reaches 124.0 Fg^{-1} at 20 mV.s^{-1}.	[This work]

metals and dyes, thereby increasing ion mobility and expanding to more applications.

Conclusion

In this study, Zn-doped aerogel (Zn-A) was successfully synthesized from sodium alginate, nipa palm shell-derived cellulose via sol–gel method combined with freeze drying. Zn-A has low density (<0.1 g/cm^3) and high porosity (more than 90 %) with a stable three-dimensional structure. Thanks to that, Zn-A has the ability to adsorb Ni^{2+} up to 194.2 mg/g corresponding to a cellulose content of 5 %. After adsorbing Ni, Zn-A was reused to create Zn, Ni-co-doped aerogel (Zn-A-Ni), then Zn-A-Ni was pyrolyzed to form carbon aerogel (Zn-CA-Ni), which has high potentials for manufacturing electrodes in super-capacitors (specific capacitance reaches 124.0F/g at 20 mV/s). In addition, with its porous carbon structure, Zn-CA-Ni also adsorbs crystal violet pigment with an adsorption capacity of 38.7 mg/g. From there, it shows that Zn-A and Zn-CA-Ni have the ability to be widely applied in the field of adsorption (heavy metals, pigments, etc.) and energy storage.

CRediT authorship contribution statement

Nguyen Huu Hieu: Methodology, Project administration. **Phan Minh Tu:** Data curation, Investigation, Writing – original draft, Writing – review & editing. **Nguyen Hoang Kim Duyen:** Investigation, Methodology. **Cao Vu Lam:** Software, Writing – original draft. **Dang Ngoc Chau Vy:** Formal analysis, Investigation, Software. **Ta Dang Khoa:** Conceptualization, Writing – review & editing. **Nguyen Truong Son:** Investigation, Methodology. **Vo Nguyen Dai Viet:** Writing – review & editing. **Pham Trong Liem Chau:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jiec.2024.07.010.

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